DBR Innovative Tech

Cultivating the Future of Agriculture with Innovation

RythuMitra

Product Name:RythuMitra

Objective of the App:To provide personalized, field-specific fertilizer recommendations based on the actual nutrient levels present in the soil.The goal is to optimize fertilizer usage, reduce unnecessary input costs, and support sustainable farming practices.

Target Users:

- Primary: Farmers (small, medium, and large-scale)

Problem Statement

Farmers in India often apply fertilizers using a blanket approach, without considering the actual nutrient condition of their soil.

This results in:

- Overuse of nitrogen fertilizers (often up to 2× the required amount)
- Soil degradation over time
- Increased input costs for farmers
- Reduced crop yield and quality

There is a clear need for a data-driven, easy-to-use solution that helps farmers make accurate fertilizer decisions at the field level.

User Persona (Primary):

Persona Name: Farmer

Age Range / Role:35–55 years, actively involved in day-to-day farming operations

User Goals:

- Receive clear, actionable fertilizer recommendations without complex data or technical terms
- Prefer simple, easy-to-follow instructions instead of numbers or detailed analysis
- Get instant guidance on:
  - What to apply
  - How much to apply
  - When to apply
- Access step-by-step actions rather than explanations or theory
- Receive daily activity recommendations and alerts (e.g., “What should I do today?”)
- Stay informed with timely reminders for farming activities (fertilizer application, irrigation, etc.)

Behavior Patterns:

- Strong preference for visual communication (icons, images, color indicators)
- Limited literacy; difficulty understanding text-heavy or English-only interfaces
- Struggles with complex navigation and multiple steps
- Responds better to audio guidance or interactive elements (e.g., voice instructions, animations)
- Looks for quick, direct solutions instead of detailed analysis
- Focused on practical outcomes, such as:
  - What to apply
  - Where to buy
  - How to use

Feature1:Soil Nutrient Analysis & Fertilizer Recommendation

This feature provides farmers with field-specific fertilizer recommendations based on real-time soil nutrient data collected through sensors.

Key Functionalities

1. Soil Nutrient Display

- Display available soil nutrients using sensor data (e.g., Nitrogen, Phosphorus, Potassium, pH, etc.)

2. Soil Health Status

- Show overall soil condition using clear indicators:
  - Healthy,Medium,Low
- Use color coding and icons for quick understanding

3. Fertilizer Recommendation Options

Provide two recommendation paths, allowing the user to choose:
Option 1: Chemical Fertilizer Recommendation

- Urea: 250 kg,DAP: 20 kg,SOP: 30 kg

Option 2: Bio Fertilizer Recommendation

- Rhizobium: 2 litres,Vermicompost

Clearly highlight that:

- Bio fertilizers are lower cost
- Environmentally safe and sustainable

4. User Choice

- Allow users to select between chemical and bio fertilizer options
- Clearly present benefits of each option, with emphasis on bio fertilizers

5. Micronutrient Recommendation (Based on Crop Type)

6. Usage Instructions

  - Easy to understand

7. Accessibility & Guidance Enhancements

- Enable voice guidance for easy listening,Show fertilizer images for quick identification

Feature 2: Crop Recommendation System

Feature Overview

This feature recommends suitable crops based on soil nutrient availability and disease resistance, helping farmers choose crops that are both profitable and sustainable for their field conditions.

Key Functionalities:

1. Recommendation Options

Option 1: Single Crop Recommendation

- Display crop cards with the following details:
  - Crop Name
  - Estimated Input Cost
  - Expected Yield per Acre
  - Estimated Profit per Acre

Option 2: Intercropping Recommendation (Multi-Crop)

- Display crop combinations with:
  - Crop Names and Yield Ratio
  - Estimated Input Cost
  - Expected Yield per Acre
  - Estimated Profit per Acre
  -

2. Crop Action (Start Cultivation)

- Each crop card should include a “Start Cultivation” button
- On selection, the crop is added to the farmer’s current crop list

3. Crop Details View

- On clicking a crop card, open a detailed view including:
  - Benefits of cultivating the crop
  - Estimated input cost
  - Expected profit
  - Additional key insights

- Include images of the crop for easy identification and better understanding

Feature 3: Soil Health Correction Plan

This feature provides a soil improvement and correction plan during non-cropping periods (e.g., summer), helping farmers restore and enhance soil fertility based on current soil conditions.

Key Functionalities

1. Seasonal Soil Correction Plan

- Provide a detailed soil correction plan during periods when no crops are being cultivated
- Recommendations are based on:
  - Current soil nutrient status
  - Soil health condition (Low / Medium / Healthy)

2. Soil Health Improvement Recommendations

- Suggest practical and low-cost methods to improve soil fertility
- Focus on easy-to-implement, farmer-friendly solutions

3. Low-Cost Home Remedies

- Recommend locally available and affordable solutions such as:
  - Organic inputs
  - Natural soil enhancers
- Ensure recommendations are simple, actionable, and cost-effective

4. Recommendations for Poor Soil Conditions

- For soils with very low nutrient levels, provide targeted solutions:
  - Vermicompost
  - Bio-nutrient boosters
- Clearly explain what to apply and how it improves soil health

5. Step-by-Step Guidance

- Provide simple, step-by-step instructions for soil correction practices
- Focus on “What to do” rather than technical explanations

Feature 4: Crop Daily Activity Planner

Feature Overview
This feature provides a day-by-day activity plan to guide farmers throughout the crop lifecycle.
It helps farmers successfully cultivate crops even without prior knowledge by offering clear, step-by-step instructions.

Key Functionalities

1. Timeline-Based Activity Planner
- Display the complete crop plan in a timeline format
- Cover the entire crop duration from Day 1 to harvest
- Ensure the flow is simple, visual, and easy to follow

2. Crop Details
- Example:
  - Crop Name: Paddy
  - Duration: 120 Days

3. Daily Activity Breakdown
Provide structured daily tasks such as:
  - Day 1:Purchase quality seeds from a certified source
  - Day 3:Soak seeds in water for 48 hours to break dormancy
- ... (Continue for all key stages)
  - Day 120:Harvest the crop
  - Dry and store properly

4. Guided Instructions
- Each activity should include:
  - Clear action steps
  - Simple language (no technical terms)

5. Smart Reminders & Alerts
- Send daily notifications for scheduled tasks
- Highlight “Today’s Task” prominently

6. Accessibility Enhancements
- Provide voice guidance for each activity
- Use icons and images to explain actions visually

Feature 5: Crop Value Addition

Feature Overview

This feature provides low-cost value addition techniques to help farmers increase profit by further processing their crops.
It is especially useful during market price drops, enabling farmers to earn better returns instead of selling raw produce.

Key Functionalities

1. Value Addition Recommendations
- Suggest simple and low-cost processing methods for harvested crops
- Focus on profit enhancement through basic transformation

2. Top 3 Recommended Options
- Display the best three value addition options for each crop
- Allow farmers to select one option based on feasibility

3. Process Details
For each recommended option, provide:
- Process Name
- How to do it (step-by-step instructions)
- Where to do it (home, local facility, or processing unit)
- Estimated Cost involved

4. Simple & Practical Guidance
- Ensure instructions are:
  - Easy to understand
  - Action-oriented
  - Suitable for farmers with limited technical knowledge

Feature 6: Voice Support

Feature Overview
This feature enables voice assistance across the entire app, making it easier for farmers to interact with the system without relying on text.
Key Functionalities
1. Voice Guidance Across the App
- Provide audio support for all major features and instructions
- Allow users to listen instead of reading